
AY2026 MA08 Applied Algebra

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Deadline: 17:00, Monday, June 22, 2026
Assignment: 2

1. a) Find $\gcd(24, 42)$.
b) Find $\gcd(437, 621)$.

Solution:

- a) Using the Euclidean algorithm,

$$42 = 1 \cdot 24 + 18,$$

$$24 = 1 \cdot 18 + 6,$$

$$18 = 3 \cdot 6 + 0.$$

Therefore,

$$\gcd(24, 42) = 6.$$

- b) Using the Euclidean algorithm,

$$621 = 1 \cdot 437 + 184,$$

$$437 = 2 \cdot 184 + 69,$$

$$184 = 2 \cdot 69 + 46,$$

$$69 = 1 \cdot 46 + 23,$$

$$46 = 2 \cdot 23 + 0.$$

Therefore,

$$\gcd(437, 621) = 23.$$

2. Determine whether the following statements are True or False and give your reasons.

- a) $17 \equiv 2 \pmod{3}$.
- b) $93 \equiv 5 \pmod{15}$.
- c) $128 \equiv 256 \pmod{8}$.

Solution:

- a) This statement is True because

$$17 - 2 = 15,$$

and 15 is divisible by 3.

b) This statement is False because

$$93 - 5 = 88,$$

and 88 is not divisible by 15.

c) This statement is True because

$$128 - 256 = -128,$$

and -128 is divisible by 8.

3. For a binary operation $*$ defined on $S = \{a, b, c\}$ by means of the following table.

$*$	a	b	c
a	c	b	a
b	a	a	a
c	b	b	b

a) Compute $a * c$, $c * b$.

b) Compute $(a * c) * b$, $a * (c * b)$. Based on this computation, can you say $*$ is associative?

c) Compute $(a * b) * c$, $a * (b * c)$. Based on this computation, can you say $*$ is associative?

d) Is $*$ commutative in this table? Why?

Solution:

a) From the table,

$$a * c = a,$$

and

$$c * b = b.$$

b) First,

$$(a * c) * b = a * b = b.$$

Also,

$$a * (c * b) = a * b = b.$$

So,

$$(a * c) * b = a * (c * b).$$

However, this computation checks only one case. Therefore, we cannot say that $*$ is associative only from this result.

c) First,

$$(a * b) * c = b * c = a.$$

Also,

$$a * (b * c) = a * a = c.$$

Therefore,

$$(a * b) * c \neq a * (b * c).$$

So, $*$ is not associative.

d) The operation $*$ is not commutative. For example,

$$a * b = b,$$

but

$$b * a = a.$$

Since

$$a * b \neq b * a,$$

the operation is not commutative.

Notice:

(1) Please write your Email title as “2026-A{assignment number}-{Your Student ID}-{Your Name}”. For example, “2026-A2-s1320037-Yuki Tahara”.